

Final Paper: Impact of Alcohol Consumption on Wages

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The relationship between alcohol consumption and labor market outcomes is a topic of social and economic interest that has widespread reach. Alcohol consumption is a global behavior, and its effects on individual productivity, career advancement, and earnings have long been debated. Some argue that moderate alcohol consumption fosters social interactions and stress relief, potentially leading to improved labor market outcomes, while others believe that excessive drinking can harm physical and mental health, reducing workplace performance and wage outcomes.

This study examines the effect of alcohol consumption on total wage and salary income, a key labor market outcome. Using data from the National Longitudinal Survey of Youth, the analysis focuses on young adults aged 24–32 in 1989 and 29–37 in 1994. The study aims to investigate whether the number of days an individual has had at least one drink in the past month is associated with differences in annual wages, controlling for other important factors such as education, age, and gender.

The findings outlined in the paper suggest that moderate alcohol consumption may have positive effects on labor market outcomes, potentially through uses such as social networking or stress management. At the same time, the results underscore the importance of addressing structural factors like education and gender equity that may have marginally larger impacts on total wage. This paper provides an overview of the methodology, results, and implications of this relationship, contributing to the broader understanding of how lifestyle choices intersect with economic outcomes.

Description of Data:

Variable	Mean	Median	Std. Dev.	Min	Max
ln(wgsal)	9.77	9.611	1.342	0	13.32
days	4.762	2	6.812	0	31
higrad	12.91	12	2.41	0	20
age	30.03	30	3.31	24	37
sex	1.516	2	0.5	1	2

Empirical Results:

This study investigates the relationship between alcohol consumption and total wage and salary income. The research question asks: What is the effect of alcohol consumption on wage and salary income? To answer this question, a multiple regression model is utilized accounting for key control factors influencing income while isolating the effect of alcohol consumption as the key independent variable.

The dependent variable in this analysis is the natural logarithm of wage and salary income. This variable provides a direct measure of labor market success and is commonly used in economic analyses due to its ability to interpret coefficients as percentage changes. Taking the logarithm of income helps smooth out large differences in wages, addressing the naturally skewed distribution of wages, allowing for a more robust and meaningful analysis.

The key independent variable is the number of days in the past month that an individual consumed alcohol. This specific variable captures the frequency of drinking and serves as a proxy for alcohol consumption behavior. This variable is particularly interesting because alcohol consumption may influence labor market outcomes through various mechanisms, such as social networking, stress management, or productivity. For example, moderate drinking might foster professional relationships and networking opportunities, potentially leading to higher wages.

Conversely, excessive drinking might have negative health or productivity effects, leading to lower earnings. Additionally, this study does not account for the context in which drinking occurs, such as whether it happens in a work or networking setting, which could influence its impact on labor market outcomes.

To isolate the effect of alcohol consumption on income, I include three control variables that are well-established determinants of wages including education, age, and gender. Education, or 'higrad', measured as the number of years of schooling completed, serves as a proxy for skill and employability. It is a critical factor influencing an individual's earnings potential. Age is included as a proxy for work experience, reflecting the typical increase in productivity and earnings as individuals spend more time in the labor force. Lastly, gender is included to control for wage disparities between men and women, which are well-documented in labor market studies. Gender, or 'sex', is represented as 1 for males and 2 for females.

Regression Model:

$$\ln(\text{wgal}) = 6.975 + 0.0064 \cdot \text{days} + 0.131 \cdot \text{higrad} + 0.052 \cdot \text{age} - 0.499 \cdot \text{sex} + \mu$$

Interpretation of Coefficients:

- Intercept (6.975) = Baseline log wage when all independent variables are 0
- β_1 : [Days] (0.0064) = Each additional day of drinking in the past month is associated with a 0.64% increase in wages.
- β_2 : [higrad] (0.131) = Each additional year of education is associated with a 13.1% increase in wages.
- β_3 : [age] (0.052) = Each additional year of age is associated with a 5.2% increase in wages.

- β_4 : [sex] (-0.499) = Being female (sex = 2) is associated with a 49.9% lower wage compared to being male.

In this model, β_1 represents the effect of alcohol consumption on log-transformed wages, holding other factors constant. The coefficients β_2 , β_3 , and β_4 capture the effects of education, age, and gender on wages, respectively, while β_0 is the intercept, and μ is the error term, accounting for unobserved factors.

The inclusion of these control variables ensures that the model isolates the relationship between alcohol consumption and wages, mitigating potential biases due to omitted variables. For example, individuals with higher education levels might also have different drinking habits, and failing to control for education could confound the relationship between alcohol consumption and wages. Similarly, gender and age are essential to consider because both influence drinking behavior and earnings.

By examining the statistical significance and magnitude of β_1 , the study evaluates whether alcohol consumption has a meaningful impact on wages and whether this relationship is positive or negative. Ultimately, the results will contribute to understanding the nuanced ways lifestyle choices, such as alcohol consumption, impact economic success.

Estimation Results:

Variable	Coefficient	Standard Error	Statistically Significant	95% CI (Lower)	95% CI (Upper)
Intercept	6.975	0.089	Yes	6.801	7.149
Days	0.0064	0.001	Yes	0.004	0.009
Education	0.131	0.004	Yes	0.124	0.138
Age	0.052	0.003	Yes	0.047	0.057
Gender (Female)	-0.499	0.017	Yes	-0.533	-0.465

Number of Observations: 16,001 R-Squared: 0.171
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The results suggest a small but statistically significant positive relationship between the frequency of alcohol consumption and wages. The variable 'days' is statistically significant, suggesting that there is strong evidence to conclude that the frequency of alcohol consumption has a measurable effect on wages. Each additional day of drinking in the past month is associated with a 0.64% increase in annual wages, even after accounting for education, age, and gender. However, this effect is modest compared to the substantial impact of education and age on earnings. Additionally, the analysis highlights persistent gender wage gaps, with women earning significantly less than men, even after controlling for observable characteristics.

Education and age are shown to have a substantial impact on wages. Both variables determining education and age were statistically significant, suggesting that there is strong evidence to conclude that each additional year in age and education have an effect on wages. Education emerges as a dominant factor, with each additional year of schooling contributing significantly to earning potential, increasing total income by 13.1% for each additional year of education. Similarly, age, as a proxy for work experience, plays a key role in wage determination, increasing total income by 5.2% for each additional year of age. These findings are consistent with economic theory, emphasizing the value of schooling and experience in the labor market.

The significant negative coefficient for gender highlights a persistent wage gap, with women earning considerably less than men, even after controlling for key factors. The variable for gender is also statistically significant, suggesting that gender has an effect on wages. The model found that being female is associated with a 49.9% lower wage compared to being male. This result points to structural issues in the labor market like discrimination, which suggests further investigation is necessary.

In this model the dependent variable is log-transformed. Using R-squared as a measure of fit can be misleading, and inaccurately explain the proportion of variation in the dependent variable explained by the model. This limitation can result in an underestimation or overestimation of the model's explanatory power. To address this issue, the pseudo-R-squared is used, which evaluates the fit of the log model relative to the variation in the dependent variable's logarithm. In this analysis, pseudo-R-squared is 0.171, suggesting that 17.1% of the variation in log-transformed wages is explained by the independent variables. While it may seem that this model is not a good fit, the modest percentage reflects the complexity of factors influencing wages.

This study examined the effect of alcohol consumption on total wage and salary, using income as a key predictor of labor market outcomes. The results of the regression analysis showed that moderate alcohol consumption, measured by the number of drinking days in the past month, has a small but statistically significant positive association with wages, suggesting a 0.64% increase in income per additional drinking day. However, the findings suggest that while alcohol consumption may play a minor role in predicting salary, factors such as education and gender are more significant predictors.